

SpaceX Falcon 9 First Stage Landing Prediction

IBM Data Science Capstone

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Executive Summary

- Goal: predict whether Falcon 9's first stage will land successfully
- Data collected from SpaceX API and Wikipedia
- EDA performed using Python, SQL, and Folium
- Interactive dashboard created with Plotly Dash
- Machine-learning models compared

Introduction

- Why first-stage landing matters
- Successful landing reduces launch cost
- Objective: identify factors related to landing success and build a prediction model

Data Collection

- SpaceX API
- Wikipedia launch-history data
- Number of Falcon 9 records
- Important variables: payload, orbit, launch site, booster version, landing outcome

Data Wrangling

- Removed unnecessary/missing data
- Converted landing outcomes into Class
- 1 = successful landing
- 0 = unsuccessful landing
- One-hot encoded categorical features
- Standardized ML features

EDA Methodology

- Pandas
- Matplotlib
- Seaborn
- Compared flight number, payload, orbit, launch site, and success rate

EDA Results

- Success rate generally increased over the years
- Later flights generally had better landing outcomes
- Show your **Payload Mass vs Flight Number** chart
- Show your **Success Rate by Orbit** chart

SQL Analysis

Include important SQL findings:

- Unique launch sites
- NASA payload mass
- Average payload for F9 v1.1
- Landing outcomes
- Maximum payload
- 2015 failed drone-ship landings

Folium Interactive Map

- Launch locations
- Successful vs unsuccessful launches
- Marker clusters
- Distance from launch site to coastline
- Explain why launch sites are located near coastlines

Plotly Dash Dashboard

Show screenshots of your dashboard:

- Launch Site dropdown
- Success/failure pie chart
- Payload Range slider
- Payload vs landing outcome scatter plot

Predictive Analysis Methodology

Models tested:

- Logistic Regression
- Support Vector Machine
- Decision Tree
- K-Nearest Neighbors

Explain:

- Train/test split = **80/20**
- Test samples = **18**
- GridSearchCV
- cv=10
- Hyperparameter tuning

Predictive Analysis Results

Model	Validation Accuracy	Test Accuracy
Logistic Regression	0.8333333333333334	83.33%
SVM	0.8333333333333334	83.33%
Decision Tree	0.8333333333333334	83.33%
KNN	0.8333333333333334	83.33%

Innovative Insights

- Landing success improved as SpaceX gained more flight experience.
- Launch site and orbit appear to influence landing outcomes.
- Payload mass alone does not determine landing success.
- Reusable booster technology becomes increasingly important in later flights.
- Historical launch data can be used to support future landing predictions.

Conclusion

- Summarize the major findings
- State which ML model performed best
- Explain how the analysis could help evaluate future Falcon 9 launches

GitHub

<https://github.com/KaungHtet6107/SpaceX-Capstone>